

Creation Date 19-Aug-2010

Revision Date 29-Sep-2023

Revision Number 8

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                      |                                                   |
|----------------------|---------------------------------------------------|
| Product Description: | <b>Dichloromethane-d2</b>                         |
| Cat No. :            | <b>321270000; 321270010; 321270050; 321270100</b> |
| CAS No               | 1665-00-5                                         |
| EC No                | 216-776-0                                         |
| Molecular Formula    | C Cl2 D2                                          |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| Recommended Use                | Laboratory chemicals.                                                                       |
| Sector of use                  | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites    |
| Product category               | PC21 - Laboratory chemicals                                                                 |
| Process categories             | PROC15 - Use as a laboratory reagent                                                        |
| Environmental release category | ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates) |
| Uses advised against           | No Information available                                                                    |

### 1.3. Details of the supplier of the safety data sheet

#### Company

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium

**UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road,  
Loughborough, Leicestershire LE11 5RG, United Kingdom

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
e-mail - infoch@thermofisher.com

**E-mail address** begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

# SAFETY DATA SHEET

Dichloromethane-d2

Revision Date 29-Sep-2023

## 2.1. Classification of the substance or mixture

### CLP Classification - Regulation (EC) No 1272/2008

#### Physical hazards

Based on available data, the classification criteria are not met

#### Health hazards

Skin Corrosion/Irritation

Category 2 (H315)

Serious Eye Damage/Eye Irritation

Category 2 (H319)

Carcinogenicity

Category 2 (H351)

Specific target organ toxicity - (single exposure)

Category 3 (H336)

#### Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Warning

### **Hazard Statements**

H315 - Causes skin irritation

H319 - Causes serious eye irritation

H336 - May cause drowsiness or dizziness

H351 - Suspected of causing cancer

### **Precautionary Statements**

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P337 + P313 - If eye irritation persists: Get medical advice/attention

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P312 - Call a POISON CENTER or doctor if you feel unwell

P280 - Wear protective gloves/protective clothing/eye protection/face protection

## 2.3. Other hazards

Contains a known or suspected endocrine disruptor

Contains a substance on the National Authorities Endocrine Disruptor Lists

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

# SAFETY DATA SHEET

Dichloromethane-d2

Revision Date 29-Sep-2023

## 3.1. Substances

| Component            | CAS No    | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                 |
|----------------------|-----------|-------------------|----------|-----------------------------------------------------------------------------------|
| Dichloro(2H2)methane | 1665-00-5 | EEC No. 216-776-0 | 100      | Skin Irrit. 2 (H315)<br>Eye Irrit. 2 (H319)<br>STOT SE 3 (H336)<br>Carc. 2 (H351) |
| Dichloromethane      | 75-09-2   | EEC No. 200-838-9 | -        | Skin Irrit. 2 (H315)<br>Eye Irrit. 2 (H319)<br>STOT SE 3 (H336)<br>Carc. 2 (H351) |

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                           |                                                                                                                   |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | If symptoms persist, call a physician.                                                                            |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Ingestion</b>                          | Clean mouth with water and drink afterwards plenty of water.                                                      |
| <b>Inhalation</b>                         | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Self-Protection of the First Aider</b> | Use personal protective equipment as required.                                                                    |

### 4.2. Most important symptoms and effects, both acute and delayed

. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

### 4.3. Indication of any immediate medical attention and special treatment needed

|                           |                                                                                                                                                                                                                                              |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Notes to Physician</b> | A patient adversely affected by exposure to this product should not be given adrenaline (epinephrine) or similar heart stimulant since these would increase the risk of cardiac arrhythmias. Treat symptomatically. Symptoms may be delayed. |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### Extinguishing media which must not be used for safety reasons

No information available.

### 5.2. Special hazards arising from the substance or mixture

# SAFETY DATA SHEET

Dichloromethane-d2

Revision Date 29-Sep-2023

Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

## Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Phosgene, Hydrogen chloride gas.

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation.

### 6.2. Environmental precautions

Should not be released into the environment.

### 6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Ensure adequate ventilation. Avoid ingestion and inhalation.

## Hygiene Measures

When using do not eat, drink or smoke. Provide regular cleaning of equipment, work area and clothing.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Store under an inert atmosphere. Protect from moisture.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 10

**Switzerland - Storage of hazardous substances**

Storage class - SC 10/12  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

# SAFETY DATA SHEET

Dichloromethane-d2

Revision Date 29-Sep-2023

## 8.1. Control parameters

### Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund). **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC

| Component       | European Union                                                                                                               | The United Kingdom                                                                                                         | France                                                                                                                                                                                                                         | Belgium                                                                                                                               | Spain                                                                                                                                                                         |
|-----------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dichloromethane | TWA: 353 mg/m <sup>3</sup> (15min)<br>TWA: 100 ppm (15min)<br>STEL: 706 mg/m <sup>3</sup> (8h)<br>STEL: 200 ppm (8h)<br>Skin | STEL: 200 ppm 15 min<br>STEL: 706 mg/m <sup>3</sup> 15 min<br>TWA: 353 mg/m <sup>3</sup> 8 hr<br>TWA: 100 ppm 8 hr<br>Skin | TWA / VME: 50 ppm (8 heures). restrictive limit<br>TWA / VME: 178 mg/m <sup>3</sup> (8 heures). restrictive limit<br>STEL / VLCT: 100 ppm. restrictive limit<br>STEL / VLCT: 356 mg/m <sup>3</sup> . restrictive limit<br>Peau | TWA: 50 ppm 8 uren<br>TWA: 177 mg/m <sup>3</sup> 8 uren<br>STEL: 200 ppm 15 minuten<br>STEL: 706 mg/m <sup>3</sup> 15 minuten<br>Huid | STEL / VLA-EC: 100 ppm (15 minutos).<br>STEL / VLA-EC: 353 mg/m <sup>3</sup> (15 minutos).<br>TWA / VLA-ED: 50 ppm (8 horas)<br>TWA / VLA-ED: 177 mg/m <sup>3</sup> (8 horas) |

| Component       | Italy                                                                                                                                                                                                          | Germany                                                                                                                                                                                                                                                              | Portugal                                                                                                                                 | The Netherlands                                                                     | Finland                                                                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dichloromethane | TWA: 175 mg/m <sup>3</sup> 8 ore.<br>Time Weighted Average<br>TWA: 50 ppm 8 ore.<br>Time Weighted Average<br>STEL: 353 mg/m <sup>3</sup> 15 minuti. Short-term<br>STEL: 100 ppm 15 minuti. Short-term<br>Pelle | TWA: 50 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 180 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 50 ppm (8 Stunden). MAK<br>TWA: 180 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 100 ppm<br>Höhepunkt: 360 mg/m <sup>3</sup><br>Haut | STEL: 706 mg/m <sup>3</sup> 15 minutos<br>STEL: 200 ppm 15 minutos<br>TWA: 353 mg/m <sup>3</sup> 8 horas<br>TWA: 100 ppm 8 horas<br>Pele | huid<br>STEL: 706 mg/m <sup>3</sup> 15 minuten<br>TWA: 353 mg/m <sup>3</sup> 8 uren | TWA: 50 ppm 8 tunteina<br>TWA: 177 mg/m <sup>3</sup> 8 tunteina<br>STEL: 100 ppm 15 minuutteina<br>STEL: 353 mg/m <sup>3</sup> 15 minuutteina<br>Iho |

| Component       | Austria                                                                                                                                                     | Denmark                                                                                                                                  | Switzerland                                                                                                                                      | Poland                                                                           | Norway                                                                                                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dichloromethane | Haut<br>MAK-KZGW: 200 ppm 15 Minuten<br>MAK-KZGW: 700 mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 50 ppm 8 Stunden<br>MAK-TMW: 175 mg/m <sup>3</sup> 8 Stunden | TWA: 35 ppm 8 timer<br>TWA: 122 mg/m <sup>3</sup> 8 timer<br>STEL: 706 mg/m <sup>3</sup> 15 minutter<br>STEL: 200 ppm 15 minutter<br>Hud | Haut/Peau<br>STEL: 200 ppm 15 Minuten<br>STEL: 706 mg/m <sup>3</sup> 15 Minuten<br>TWA: 50 ppm 8 Stunden<br>TWA: 177 mg/m <sup>3</sup> 8 Stunden | STEL: 353 mg/m <sup>3</sup> 15 minutach<br>TWA: 88 mg/m <sup>3</sup> 8 godzinach | TWA: 15 ppm 8 timer<br>TWA: 50 mg/m <sup>3</sup> 8 timer<br>STEL: 45 ppm 15 minutter. value from the regulation<br>STEL: 150 mg/m <sup>3</sup> 15 minutter. value from the regulation<br>Hud |

| Component       | Bulgaria                                                                                                      | Croatia                                                                                                                                                            | Ireland                                                                                                                      | Cyprus                                                                                                                                | Czech Republic                                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Dichloromethane | TWA: 353 mg/m <sup>3</sup><br>TWA: 100 ppm<br>STEL : 706 mg/m <sup>3</sup><br>STEL : 200 ppm<br>Skin notation | kože<br>TWA-GVI: 100 ppm 8 satima.<br>TWA-GVI: 353 mg/m <sup>3</sup> 8 satima.<br>STEL-KGVI: 200 ppm 15 minutama.<br>STEL-KGVI: 706 mg/m <sup>3</sup> 15 minutama. | TWA: 100 ppm 8 hr.<br>TWA: 353 mg/m <sup>3</sup> 8 hr.<br>STEL: 200 ppm 15 min<br>STEL: 706 mg/m <sup>3</sup> 15 min<br>Skin | Skin-potential for cutaneous absorption<br>STEL: 706 mg/m <sup>3</sup><br>STEL: 200 ppm<br>TWA: 353 mg/m <sup>3</sup><br>TWA: 100 ppm | TWA: 200 mg/m <sup>3</sup> 8 hodinách.<br>Potential for cutaneous absorption<br>Ceiling: 500 mg/m <sup>3</sup> |

| Component       | Estonia                                                         | Gibraltar                                                                                               | Greece                                                                                    | Hungary                                                                                 | Iceland                                                                     |
|-----------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Dichloromethane | Nahk<br>TWA: 35 ppm 8 tundides.<br>TWA: 120 mg/m <sup>3</sup> 8 | Skin notation<br>TWA: 353 mg/m <sup>3</sup> 8 hr<br>TWA: 100 ppm 8 hr<br>STEL: 706 mg/m <sup>3</sup> 15 | skin - potential for cutaneous absorption<br>STEL: 200 ppm<br>STEL: 706 mg/m <sup>3</sup> | STEL: 706 mg/m <sup>3</sup> 15 percekben. CK<br>TWA: 353 mg/m <sup>3</sup> 8 órában. AK | TWA: 35 ppm 8 klukkustundum.<br>TWA: 122 mg/m <sup>3</sup> 8 klukkustundum. |

# SAFETY DATA SHEET

Dichloromethane-d2

Revision Date 29-Sep-2023

|  |                                                                                            |                             |                                            |                                            |                                                                    |
|--|--------------------------------------------------------------------------------------------|-----------------------------|--------------------------------------------|--------------------------------------------|--------------------------------------------------------------------|
|  | tundides.<br>STEL: 70 ppm 15<br>minutites.<br>STEL: 250 mg/m <sup>3</sup> 15<br>minutites. | min<br>STEL: 200 ppm 15 min | TWA: 100 ppm<br>TWA: 353 mg/m <sup>3</sup> | lehetséges borön<br>keresztüli felszívódás | Skin notation<br>Ceiling: 70 ppm<br>Ceiling: 244 mg/m <sup>3</sup> |
|--|--------------------------------------------------------------------------------------------|-----------------------------|--------------------------------------------|--------------------------------------------|--------------------------------------------------------------------|

| Component       | Latvia                                                                                                                                 | Lithuania                                                                                                 | Luxembourg                                                                                                                                                                                                | Malta                                                                                                                                                                         | Romania                                                                                                                                           |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Dichloromethane | skin - potential for<br>cutaneous exposure<br>STEL: 150 mg/m <sup>3</sup><br>STEL: 42 ppm<br>TWA: 120 mg/m <sup>3</sup><br>TWA: 34 ppm | TWA: 35 ppm IPRD<br>TWA: 120 mg/m <sup>3</sup> IPRD<br>Oda<br>STEL: 70 ppm<br>STEL: 250 mg/m <sup>3</sup> | Possibility of significant<br>uptake through the skin<br>TWA: 100 ppm 8<br>Stunden<br>TWA: 353 mg/m <sup>3</sup> 8<br>Stunden<br>STEL: 200 ppm 15<br>Minuten<br>STEL: 706 mg/m <sup>3</sup> 15<br>Minuten | possibility of significant<br>uptake through the skin<br>TWA: 100 ppm<br>TWA: 353 mg/m <sup>3</sup><br>STEL: 200 ppm 15<br>minuti<br>STEL: 706 mg/m <sup>3</sup> 15<br>minuti | Skin notation<br>TWA: 100 ppm 8 ore<br>TWA: 353 mg/m <sup>3</sup> 8 ore<br>STEL: 200 ppm 15<br>minute<br>STEL: 706 mg/m <sup>3</sup> 15<br>minute |

| Component       | Russia                                                       | Slovak Republic                                                                                                       | Slovenia                                                                                                                                     | Sweden                                                                                                                                                                        | Turkey |
|-----------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Dichloromethane | TWA: 50 mg/m <sup>3</sup> 0922<br>MAC: 100 mg/m <sup>3</sup> | Ceiling: 706 mg/m <sup>3</sup><br>Potential for cutaneous<br>absorption<br>TWA: 100 ppm<br>TWA: 353 mg/m <sup>3</sup> | TWA: 100 ppm 8 urah<br>TWA: 353 mg/m <sup>3</sup> 8 urah<br>Koža<br>STEL: 200 ppm 15<br>minutah<br>STEL: 706 mg/m <sup>3</sup> 15<br>minutah | Binding STEL: 70 ppm<br>15 minuter<br>Binding STEL: 250<br>mg/m <sup>3</sup> 15 minuter<br>TLV: 35 ppm 8 timmar.<br>NGV<br>TLV: 120 mg/m <sup>3</sup> 8<br>timmar. NGV<br>Hud |        |

## Biological limit values

List source(s): **UK** - Biological Monitoring Guidance Values provided by the UK's Health and Safety Executive (HSE) Control of Substances Hazardous to Health Regulations (COSHH) 2002 (as amended) and EH40/2005.

| Component       | European Union | United Kingdom                                            | France                                                                                                         | Spain                                           | Germany                                                                      |
|-----------------|----------------|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------|------------------------------------------------------------------------------|
| Dichloromethane |                | Carbon monoxide: 30<br>ppm end-tidal breath<br>post shift | Dichloromethane: 0.2<br>mg/L urine end of shift<br>Carboxyhémoglobine<br>sanguine: 3.5 % blood<br>end of shift | Dichloromethane: 0.3<br>mg/L urine end of shift | Dichloromethane: 500<br>µg/L whole blood<br>(immediately after<br>exposure ) |

| Component       | Italy | Finland | Denmark | Bulgaria | Romania                                                                                                                                                              |
|-----------------|-------|---------|---------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dichloromethane |       |         |         |          | Carboxyhemoglobin: 5<br>% Hemoglobin blood<br>end of shift<br>Methylene chloride: 0.3<br>mg/L urine end of shift<br>Methylene chloride: 1<br>mg/L blood end of shift |

| Component       | Gibraltar | Latvia | Slovak Republic                                                                                                                                        | Luxembourg | Turkey |
|-----------------|-----------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------|
| Dichloromethane |           |        | Dichloromethane: 1<br>mg/L blood end of<br>exposure or work shift<br>Carboxyhemoglobin: 5<br>% of hemoglobin blood<br>end of exposure or work<br>shift |            |        |

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

# SAFETY DATA SHEET

Dichloromethane-d2

Revision Date 29-Sep-2023

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                      | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|--------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Dichloromethane<br>75-09-2 (-) |                              |                                 |                                | DNEL = 12mg/kg<br>bw/day          |

| Component                      | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|--------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Dichloromethane<br>75-09-2 (-) |                                  | DMEL = 132.14mg/m <sup>3</sup>      |                                    | DNEL = 176mg/m <sup>3</sup>           |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                      | Fresh water                       | Fresh water sediment                                              | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)                                        |
|--------------------------------|-----------------------------------|-------------------------------------------------------------------|--------------------|------------------------------------|-----------------------------------------------------------|
| Dichloromethane<br>75-09-2 (-) | PNEC = 130µg/L<br>PNEC = 0.31mg/L | PNEC = 163µg/kg<br>sediment dw<br>PNEC = 2.57mg/kg<br>sediment dw | PNEC = 0.27mg/L    | PNEC = 26mg/L                      | PNEC = 173µg/kg<br>soil dw<br>PNEC = 0.33mg/kg<br>soil dw |

| Component                      | Marine water                       | Marine water sediment                                             | Marine water Intermittent | Food chain | Air |
|--------------------------------|------------------------------------|-------------------------------------------------------------------|---------------------------|------------|-----|
| Dichloromethane<br>75-09-2 (-) | PNEC = 130µg/L<br>PNEC = 0.031mg/L | PNEC = 163µg/kg<br>sediment dw<br>PNEC = 0.26mg/kg<br>sediment dw | PNEC = 0.027mg/L          |            |     |

## 8.2. Exposure controls

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Viton (R)      | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used

# SAFETY DATA SHEET

Dichloromethane-d2

Revision Date 29-Sep-2023

and maintained properly

|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Large scale/emergency use</b>       | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br><b>Recommended Filter type:</b> low boiling organic solvent Type AX Brown conforming to EN371                                                                        |
| <b>Small scale/Laboratory use</b>      | Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.<br><b>Recommended half mask:-</b> Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141<br>When RPE is used a face piece Fit Test should be conducted |
| <b>Environmental exposure controls</b> | Do not allow material to contaminate ground water system.                                                                                                                                                                                                                                                                   |

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                                |                                              |                                          |
|------------------------------------------------|----------------------------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                                       |                                          |
| <b>Appearance</b>                              | Colorless                                    |                                          |
| <b>Odor</b>                                    | sweet                                        |                                          |
| <b>Odor Threshold</b>                          | No data available                            |                                          |
| <b>Melting Point/Range</b>                     | -97 °C / -142.6 °F                           |                                          |
| <b>Softening Point</b>                         | No data available                            |                                          |
| <b>Boiling Point/Range</b>                     | 40 °C / 104 °F                               | @ 760 mmHg                               |
| <b>Flammability (liquid)</b>                   | No data available                            |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable                               | Liquid                                   |
| <b>Explosion Limits</b>                        | <b>Lower</b> 13 Vol%<br><b>Upper</b> 22 Vol% |                                          |
| <b>Flash Point</b>                             | No information available                     | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | 556 °C / 1032.8 °F                           |                                          |
| <b>Decomposition Temperature</b>               | 120 °C                                       |                                          |
| <b>pH</b>                                      | No information available                     |                                          |
| <b>Viscosity</b>                               | No data available                            |                                          |
| <b>Water Solubility</b>                        | Insoluble                                    |                                          |
| <b>Solubility in other solvents</b>            | No information available                     |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                              |                                          |
| <b>Component</b>                               | <b>log Pow</b>                               |                                          |
| Dichloromethane                                | 1.25                                         |                                          |
| <b>Vapor Pressure</b>                          | 450 hPa @ 20 °C                              |                                          |
| <b>Density / Specific Gravity</b>              | 1.360                                        |                                          |
| <b>Bulk Density</b>                            | Not applicable                               | Liquid                                   |
| <b>Vapor Density</b>                           | No data available                            | (Air = 1.0)                              |
| <b>Particle characteristics</b>                | Not applicable (liquid)                      |                                          |

### 9.2. Other information

|                          |          |
|--------------------------|----------|
| <b>Molecular Formula</b> | C Cl2 D2 |
| <b>Molecular Weight</b>  | 86.95    |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available



# SAFETY DATA SHEET

Dichloromethane-d2

Revision Date 29-Sep-2023

## 10.2. Chemical stability

Hygroscopic.

## 10.3. Possibility of hazardous reactions

### Hazardous Polymerization Hazardous Reactions

Hazardous polymerization does not occur.  
None under normal processing.

## 10.4. Conditions to avoid

Incompatible products. Excess heat. Exposure to moist air or water.

## 10.5. Incompatible materials

Strong oxidizing agents. Strong acids. Amines.

## 10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Phosgene. Hydrogen chloride gas.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

| Component       | LD50 Oral            | LD50 Dermal          | LC50 Inhalation                                            |
|-----------------|----------------------|----------------------|------------------------------------------------------------|
| Dichloromethane | > 2000 mg/kg ( Rat ) | > 2000 mg/kg ( Rat ) | 53 mg/L ( Rat ) 6 h<br>76000 mg/m <sup>3</sup> ( Rat ) 4 h |

#### (b) skin corrosion/irritation;

Category 2

#### (c) serious eye damage/irritation;

Category 2

#### (d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Based on available data, the classification criteria are not met

#### (e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

#### (f) carcinogenicity;

Category 2

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component       | EU | UK | Germany | IARC     |
|-----------------|----|----|---------|----------|
| Dichloromethane |    |    |         | Group 2A |

#### (g) reproductive toxicity;

Based on available data, the classification criteria are not met

#### (h) STOT-single exposure;

Category 3

Results / Target organs

Central nervous system (CNS).

# SAFETY DATA SHEET

Dichloromethane-d2

Revision Date 29-Sep-2023

|                                            |                                                                                                                      |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| (i) STOT-repeated exposure;                | Based on available data, the classification criteria are not met                                                     |
| Target Organs                              | None known.                                                                                                          |
| (j) aspiration hazard;                     | Based on available data, the classification criteria are not met                                                     |
| Other Adverse Effects                      | Tumorigenic effects have been reported in experimental animals.                                                      |
| Symptoms / effects, both acute and delayed | Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. |

## 11.2. Information on other hazards

|                                                                                               |                                                                                 |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Endocrine Disrupting Properties<br>Assess endocrine disrupting<br>properties for human health | .<br>Contains a substance on the National Authorities Endocrine Disruptor Lists |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

Ecotoxicity effects This product contains the following substance(s) which are hazardous for the environment.

| Component            | Freshwater Fish                        | Water Flea         | Freshwater Algae   |
|----------------------|----------------------------------------|--------------------|--------------------|
| Dichloro(2H2)methane | Pimephales promelas: LC50:193 mg/L/96h | EC50: 140 mg/L/48h | EC50:>660 mg/L/96h |
| Dichloromethane      | Pimephales promelas: LC50:193 mg/L/96h | EC50: 140 mg/L/48h | EC50:>660 mg/L/96h |

| Component            | Microtox                                    | M-Factor |
|----------------------|---------------------------------------------|----------|
| Dichloro(2H2)methane | EC50: 1 mg/L/24 h<br>EC50: 2.88 mg/L/15 min |          |
| Dichloromethane      | EC50: 1 mg/L/24 h<br>EC50: 2.88 mg/L/15 min |          |

### 12.2. Persistence and degradability

Persistence Persistence is unlikely, based on information available.

### 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

| Component       | log Pow | Bioconcentration factor (BCF) |
|-----------------|---------|-------------------------------|
| Dichloromethane | 1.25    | 6.4 - 40 dimensionless        |

### 12.4. Mobility in soil

The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air.

### 12.5. Results of PBT and vPvB assessment

No data available for assessment.

### 12.6. Endocrine disrupting properties

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

# SAFETY DATA SHEET

Dichloromethane-d2

Revision Date 29-Sep-2023

## 12.7. Other adverse effects

**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

**Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with federal, state and local regulations. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Do not reuse empty containers. Dispose of in accordance with local regulations. Dispose of this container to hazardous or special waste collection point.

**European Waste Catalogue (EWC)**

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

**EWC Waste Disposal No**  
**Other Information**

No data available  
Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

**Switzerland - Waste Ordinance**

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

**14.1. UN number**

UN1593

**14.2. UN proper shipping name**

DICHLOROMETHANE

**14.3. Transport hazard class(es)**

6.1

**14.4. Packing group**

III

### ADR

**14.1. UN number**

UN1593

**14.2. UN proper shipping name**

DICHLOROMETHANE

**14.3. Transport hazard class(es)**

6.1

**14.4. Packing group**

III

### IATA

**14.1. UN number**

UN1593

**14.2. UN proper shipping name**

DICHLOROMETHANE

**14.3. Transport hazard class(es)**

6.1

**14.4. Packing group**

III

**14.5. Environmental hazards**

No hazards identified

**14.6. Special precautions for user**

No special precautions required.

**14.7. Maritime transport in bulk according to IMO instruments**

Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

# SAFETY DATA SHEET

Dichloromethane-d2

Revision Date 29-Sep-2023

## 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component            | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|----------------------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| Dichloro(2H2)methane | 1665-00-5 | 216-776-0 | -      | -   | -     | X    | -        | -    | -    |
| Dichloromethane      | 75-09-2   | 200-838-9 | -      | -   | X     | X    | KE-23893 | X    | X    |

| Component            | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|----------------------|-----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Dichloro(2H2)methane | 1665-00-5 | -    | -                                             | -   | -    | -    | X     | -     |
| Dichloromethane      | 75-09-2   | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### Authorisation/Restrictions according to EU REACH

| Component            | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|----------------------|-----------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Dichloro(2H2)methane | 1665-00-5 | -                                                                   | -                                                                                                                                        | -                                                                                                     |
| Dichloromethane      | 75-09-2   | -                                                                   | Use restricted. See item 59.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | -                                                                                                     |

### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

### Seveso III Directive (2012/18/EC)

| Component            | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|----------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Dichloro(2H2)methane | 1665-00-5 | Not applicable                                                                            | Not applicable                                                                           |
| Dichloromethane      | 75-09-2   | Not applicable                                                                            | Not applicable                                                                           |

### Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

### Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

# SAFETY DATA SHEET

Dichloromethane-d2

Revision Date 29-Sep-2023

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

**WGK Classification** See table for values

| Component       | Germany - Water Classification (AwSV) | Germany - TA-Luft Class                              |
|-----------------|---------------------------------------|------------------------------------------------------|
| Dichloromethane | WGK2                                  | Class I : 20 mg/m <sup>3</sup> (Massenkonzentration) |

| Component       | France - INRS (Tables of occupational diseases)      |
|-----------------|------------------------------------------------------|
| Dichloromethane | Tableaux des maladies professionnelles (TMP) - RG 12 |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                        | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|----------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Dichloromethane<br>75-09-2 ( - ) | Persistent Organic Pollutants (POPs)<br>Prohibited and Restricted Substances                                   | Group I                                                                         |                                                                                             |

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H315 - Causes skin irritation

H319 - Causes serious eye irritation

H336 - May cause drowsiness or dizziness

H351 - Suspected of causing cancer

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

# SAFETY DATA SHEET

Dichloromethane-d2

Revision Date 29-Sep-2023

Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**OECD** - Organisation for Economic Co-operation and Development

**ATE** - Acute Toxicity Estimate

**BCF** - Bioconcentration factor

**VOC** - (volatile organic compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

**Creation Date** 19-Aug-2010

**Revision Date** 29-Sep-2023

**Revision Summary** Not applicable.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).**

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**